

From Ultraclean Graphene to Cosmological Analogies: Exploring Dirac Fluids Across Scales (TU_GUT_SYSY v24.1)

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Abstract

Recent experiments on ultraclean graphene have revealed collective electron behavior resembling a Dirac fluid, characterized by near-perfect fluidity, strong violation of the Wiedemann-Franz law, and thermal shear viscosity approaching fundamental quantum bounds. These findings provide a laboratory analog for strongly interacting relativistic quantum systems, including the quark-gluon plasma and holographic models. We review these developments and cautiously explore potential structural analogies to cosmological scales, such as local dark matter density estimates and low-viscosity fluids in early-universe scenarios. No direct unification is claimed; rather, the shared mathematical framework of relativistic hydrodynamics suggests avenues for future theoretical investigation.

1 The Dirac Fluid in Ultraclean Graphene

In graphene near the charge-neutrality point, electrons behave as massless Dirac fermions with Fermi velocity

$$v_F \approx 1 \times 10^6 \text{ m/s}$$

(long established, e.g., Novoselov et al., 2005)[2].

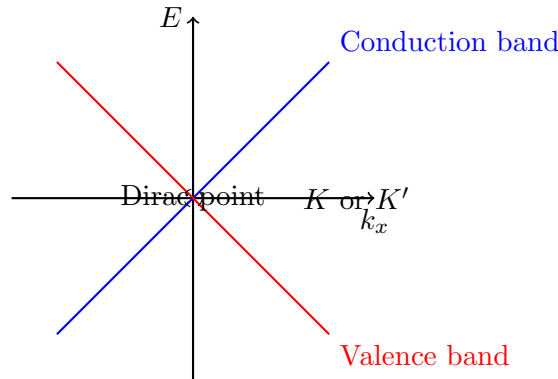


Figure 1: Schematic of the Dirac cone in graphene: linear dispersion relation $E = \pm\hbar v_F |k|$ near the Dirac point.

Recent advances in encapsulated high-mobility devices have enabled observation of hydrodynamic flow. Key results from Majumdar et al. (Nature Physics, 2025) include a giant violation of the Wiedemann-Franz law and thermal shear viscosity approaching the holographic bound

$$\frac{\eta_{\text{th}}}{s_{\text{th}}} \approx \frac{\hbar}{4\pi k_B}$$

within a factor of ~ 4 near room temperature (see Fig. 2)[1].

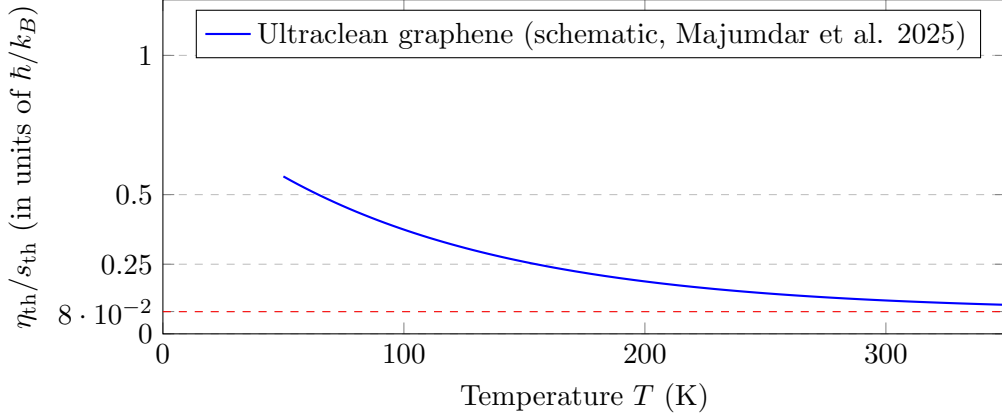


Figure 2: Schematic plot of thermal viscosity-to-entropy ratio in ultraclean graphene, approaching the Kovtun-Son-Starinets (KSS) bound at higher temperatures.

2 The Kovtun-Son-Starinets (KSS) Bound and Applications in Relativistic Fluid Dynamics

The Kovtun-Son-Starinets (KSS) bound[3] conjectures a universal lower limit on the shear viscosity-to-entropy density ratio for strongly interacting quantum fluids:

$$\frac{\eta}{s} \geq \frac{\hbar}{4\pi k_B} \approx 0.08 \frac{\hbar}{k_B}$$

derived from holographic AdS/CFT duality and uncertainty-principle arguments. Real-world applications include:

- **Quark-Gluon Plasma (QGP):** Created in heavy-ion collisions (RHIC/LHC), the QGP exhibits $\eta/s \approx 0.1 - 0.2$ (in units of $\hbar/k_B$), the closest known system to the bound[6].
- **Ultraclean Graphene Dirac Fluid:** Near the charge-neutrality point, experiments show $\eta/s \approx 0.3 - 0.8$ near room temperature[1, 7, 8], approaching the KSS bound within a factor of $\sim 4-10$.
- **Unitary Fermi Gases:** Cold atoms near unitarity reach $\eta/s \approx 0.4 - 0.5$ [9].

These low values highlight universal hydrodynamic behavior in strongly coupled relativistic systems across vastly different energy scales.

3 Analogies to High-Energy and Gravitational Physics

The low viscosity and relativistic dispersion mirror properties of the quark-gluon plasma ($\eta/s \approx 1/(4\pi)$) and holographic AdS/CFT duals[3].

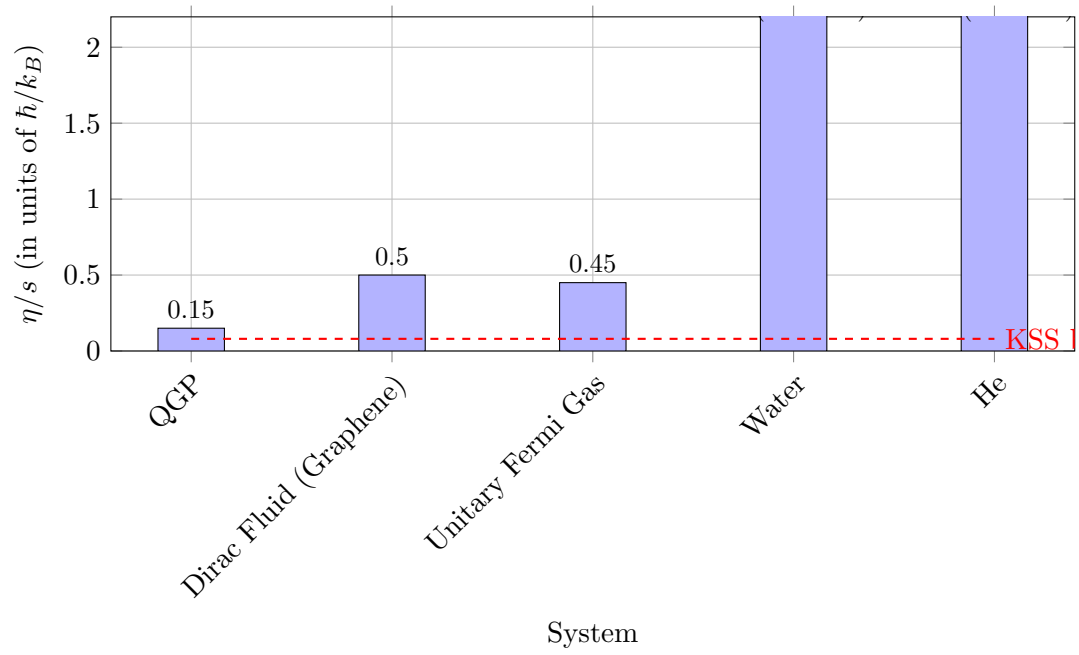


Figure 3: Comparison of η/s values across systems (approximate experimental/theoretical averages). The Dirac fluid in graphene approaches the KSS bound, similar to QGP. High values for classical fluids are off-scale.

4 Speculative Cosmological Connections and Topological Analogies

This work extends ideas presented in a previous version [14]. <https://doi.org/10.5281/zenodo.17932055>

Observed local dark matter density lies in the range $\rho_{\text{DM}} \approx 0.3 - 0.6 \text{ GeV/cm}^3$ [4, 10], while galactic magnetic field coherence scales span $\sim 1\text{--}10 \text{ kpc}$ [5, 11].

Representative values toward the lower end of these ranges (e.g., $\rho_{\text{DM}} \sim 0.3 \text{ GeV/cm}^3$, $R_{\text{coh}} \sim 1 \text{ kpc}$) align with structural analogies to the ultraclean, near-dissipationless regime of the graphene Dirac fluid [1], where hydrodynamic flow exhibits minimal turbulence and viscosity approaching quantum bounds.

In this ideal limit, topological invariants such as helicity and vortex linking are conserved with high fidelity [12, 13], evoking speculative parallels to protected braiding in anyonic systems. While no direct emergence of exotic statistics from classical 3D turbulence is established, such analogies highlight universal features of low-dissipation relativistic fluids across scales—from laboratory graphene to cosmological structures.

Future ultra-clean graphene experiments (projected zero-puddle regimes) may further illuminate these connections through enhanced topological protection in electron flow.

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